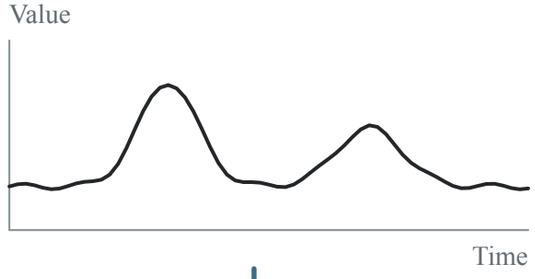
1 Retrieve references

Target sequence x



Similarity shortlist

LLM relevance review

Selected neighbor



Reference set R

2 Construct positive views

Choose a transformation using x and R

Jittering · Time masking · Permutation

Example candidates

A **Small jitter**



B **Block permutation**



Same time axis

3 Review each view

Compare with x

LLM judgment + rationale

✓ **"Correct"**

Retain in memory M

✗ **"Error"**

Refine retrieval

Rejected view: repeat reference selection

Contrastive learning

x + **M ∪ R**

Positive samples

→ **Shared encoder** →

Learn representations

Bring positive pairs together